ViDBpedia: An Inference-Rich Vietnamese DBpedia from Wikipedia Infoboxes

Le Phu Tai
20252268M
SoICT, HUST

Nguyen Thi Mai Anh
20252265M
SoICT, HUST

Duong Hoang Hai
20252273M
SoICT, HUST

Le Van Luan
20252572M
SoICT, HUST

Abstract

Knowledge specific to Vietnam (its dynasties, mandarins, battles and the administrative map redrawn in 2025) is thinly covered by DBpedia, and much of it is never written in any article. We build **ViDBpedia**, a Vietnamese DBpedia, from the official Vietnamese Wikipedia dumps, and ask whether *the ontology can make the knowledge graph contain more than was extracted*. At its core is `vidbo`, an OWL 2 DL ontology of 48 classes and 115 properties designed from 18 competency questions: 13 defined classes classify entities by their relations (a monarch is a person who holds a reign), and 10 property chains derive kinship, dynastic membership, geographic containment and war participation. A dump-based pipeline extracts 1.51M triples from 44,003 articles, and a generic OWL 2 RL reasoner plus six SPARQL rules materialises 2.64M more. Our experiments show that (i) all 18 competency questions are answered (median 44 ms), each relying on inferred triples; (ii) the same axioms and SHACL shapes expose about 1,700 data-quality issues, including bugs in our own extractor; (iii) links to English DBpedia combine 32,136 verified interlanguage links with a Silk-style matcher of 96.4% precision and 48.5% recall; and (iv) the HermiT tableau reasoner confirms our engine's inferences on a sample. We also report where the approach fails, and publish the graph through a SPARQL endpoint, Linked Data pages and a command-line client.

1 Introduction

Motivation. Wikipedia is written for people. The infobox of the Vietnamese article on Emperor Trần Nhân Tông states that his father was Trần Thánh Tông, that he reigned from 1278 to 1293 and, in the article on the Battle of Bạch Đằng, that he was one of its commanders. A machine cannot use any of this: it is wiki-text, not data. And even a person reading the page cannot answer the simple question "who was his paternal grandfather?" without opening a second article. DBpedia [1; 12] solved the first half of this problem for English by extracting infobox content into RDF and publishing it as Linked Data [2; 9]; language chapters such as the Greek edition [11] showed that local knowledge benefits from a local extraction. Vietnamese Wikipedia has about 1.6 million articles, yet the English DBpedia contains few Vietnamese dynasties, mandarins and battles, and nothing about the 2025 merger of provinces.

Why it is hard. Three properties make a Vietnamese DBpedia more than a copy of the English pipeline. First, **heterogeneous input**: the same infobox appears under more than 100 template names, with Vietnamese and English parameter names, and values such as "14 tháng 4, năm 2021", "257 TCN" (257 BCE) or "4.412.264 người" that no off-the-shelf parser reads. Second, and more importantly, **implicit knowledge**: the answers to most interesting questions are spread over several articles (a grandfather is in the father's article; a person born in a commune was born within its province; a commander of a battle took part in the enclosing war). Third, **roles and change**: monarch, mandarin and general are roles a person plays for a period, the royalty

infobox is shared by monarchs, consorts and heirs, and the administrative map changed in 2025, when Vietnam merged its provinces into 34 and abolished the district level [14].

Our approach. We follow the DBpedia architecture (dumps, mappings, extraction, linking, publication) but put an expressive ontology at its centre and *materialise* its consequences. Each competency question that the graph should answer is turned into a SPARQL query, and axioms are added until the query is answered from the data: defined classes for roles, property chains for multi-article relations, transitive and inverse properties for hierarchies, reified tenures for time-bound roles, and explicit domain knowledge about the 2025 reform. The ontology is used twice: during extraction, to reject triples whose types contradict a property's domain or range, and during reasoning, to derive new triples and to detect contradictions.

Contributions.

1. **`vidbo`**, an OWL 2 DL ontology derived from 18 competency questions, whose defined classes, property chains, transitive and inverse properties infer kinship, dynastic membership, roles, geographic containment and war participation; it models reigns and offices as n-ary relations and encodes the 2025 administrative reform.
2. A **reproducible dump-based extraction pipeline** with declarative infobox mappings, Vietnamese value parsers and a DBpedia-style ontology range check, producing 1.51M triples from 44,003 articles.
3. A **generic OWL 2 RL forward-chaining reasoner** with a SPARQL rule layer that materialises 2.64M triples in 15 minutes on a laptop, cross-checked against the HermiT tableau reasoner.
4. **Two-stage interlinking** with English DBpedia, evaluated against interlanguage links as a gold standard, and a SPARQL endpoint with dereferenceable Linked Data pages and a terminal client.
5. An **analysis of failure cases**: where linking fails, how axioms act as a bug detector, and which of our own modelling decisions produced millions of wrong triples before we fixed them.

2 Related Work

DBpedia and its chapters. DBpedia extracts Wikipedia dumps with a framework of extractors; the mapping-based extractor turns infobox parameters into properties of the community-maintained DBpedia ontology, guided by a mappings wiki [3; 12]. Internationalised chapters use IRIs in the local language and interlanguage links to connect to the English edition [11]. Wikidata [20] and YAGO [17] are complementary knowledge graphs; YAGO in particular combines Wikipedia with a taxonomy to type entities. Our work reuses the DBpedia architecture and naming conventions, but replaces the shallow DBpedia ontology for our domain by a more expressive one and materialises its consequences.

Linked Data and its quality. The Linked Data principles and the 5-star scheme [2] require data to be published in open, machine-readable formats, identified by dereferenceable URIs, queryable with W3C standards such as SPARQL 1.1 [8], and linked to other datasets. Knowledge-graph refinement and Linked Data quality are surveyed in [16; 22]; we use two of the refinement strategies they describe, axiom-based error detection and type inference, inside the construction pipeline rather than afterwards.

Ontology engineering. We follow Grüninger and Fox [6]: informal competency questions (CQs) are formalised as queries and drive the choice of axioms. OntoClean [7] distinguishes rigid from non-rigid classes, which is why roles are defined rather than asserted in `vidbo`, and ontology design patterns [4], in particular the W3C n-ary relation pattern [15], guide the modelling of reigns and offices.

Reasoning and validation. OWL 2 [21] offers several profiles [13]; OWL 2 RL can be implemented by forward-chaining rules and scales to large ABoxes, whereas tableau reasoners such as HermiT [5] decide full OWL 2 DL but do not scale to millions of assertions. We use both: an RL engine on the full data and HermiT on a sample. SHACL [10] adds closed-world validation, complementing the open-world semantics of OWL.

Link discovery. Silk [19] specifies linkage rules that combine similarity metrics with aggregations and thresholds. Our matcher implements such a rule (label similarity, type compatibility and dates) and we also provide it as a Silk link specification.

3 Problem Formulation, Data and Task

Formulation. Let W be a set of Wikipedia articles and $\mathcal{O}$ an ontology. Extraction is a function that maps each article to a set of RDF triples using a mapping M from infobox parameters to ontology terms,

$$\mathcal{A} \;=\; \bigcup_{w \in W} \mathrm{ext}_M(w) \;\subseteq\; I \times P \times (I \cup L), \tag{1}$$

where I is the set of resource IRIs, P the properties and L the literals. Reasoning computes the closure of the extracted triples under the axioms of $\mathcal{O}$ and a set of rules $\mathcal{R}$; the inferred part is what the ontology adds,

$$\mathcal{A}^* = \mathrm{Cl}_{\mathcal{O},\mathcal{R}}(\mathcal{A}), \qquad \mathcal{A}_{\mathrm{inf}} = \mathcal{A}^* \setminus \mathcal{A}. \tag{2}$$

Linking is a partial function $\ell : I \rightharpoonup I_{\mathrm{DBpedia}}$ published as `owl:sameAs`. The task is judged by whether the competency questions, written as SPARQL queries $q_1, \ldots, q_{18}$, return correct answers over $\mathcal{A}^*$, by the size and correctness of $\mathcal{A}_{\mathrm{inf}}$, and by the precision and recall of ℓ.

Data. We use the official Vietnamese Wikipedia dumps of 1 October 2026: the 1.16 GB articles dump, the language-link table and the page-properties table (Wikidata QIDs and disambiguation flags). A first pass over all 1,622,356 articles keeps those that use one of the mapped infobox templates or belong to seed categories such as *Vietnamese dynasties*, and records 317,604 redirects. A second pass adds pages linked from object-valued infobox fields (parents, birth places, commanders), so that the objects of extracted triples are themselves described. The result is 44,003 articles, 43,614 of which are not disambiguation pages. The most frequent infobox groups are person (11,473 articles), officeholder (7,134), royalty (7,078), Vietnamese administrative unit (4,170), military conflict (2,366), country or former state (1,754), settlement (1,446), military person (1,332), writer (1,074), feudal-era official (560) and book (89). From them the extraction reads 209,775 object values (links), 70,865 literals, 43,873 dates and 11,205 numbers, and creates 20,755 office, 4,691 reign and 1,807 royal-title nodes.

The task as competency questions. The usage scenario is a user who queries Vietnamese history, geography and notable people with SPARQL. We wrote 18 informal CQs before designing the ontology (Table 7 in Section 6); examples are “who is the paternal grandfather of Trần Nhân Tông?”, “who was born within today’s Nghệ An province?”, “which monarchs reigned while their dynasty was at war?” and “which districts were abolished on 1 July 2025?”. None of them can be answered by a single infobox.

4 Method

Figure 1 summarises the pipeline. The ontology drives both the extraction (range check) and the reasoning; one command rebuilds everything in about 1.5 hours on a laptop.

4.1 The `vidbo` ontology

From questions to terminology. Terminology was grounded in the data: counting how often every infobox parameter is used revealed, e.g., that 492 articles use the variant `Infobox officeholder 1` and 555 a dedicated template for feudal-era officials.

Class hierarchy. Figure 2 shows the 48 classes under five top-level classes *Agent*, *Place*, *Event*, *Work* and *Tenure*, aligned with DBpedia, FOAF and schema.org by 53 equivalence and subsumption axioms, so that the graph can also be queried with those vocabularies. Disjointness axioms ($\mathit{Person} \sqcap \mathit{Place} \sqsubseteq \bot$, eight `AllDisjointClasses` groups in total) do not add facts but expose contradictions. *Group* (dynasties) is deliberately *not* disjoint with *Place*: Wikipedia uses one article, e.g. “Nhà Nguyên”, for both a ruling house and the polity it ruled.

Roles as defined classes. Being a monarch, an official or a commander is a *role* that a person plays for a period, not a rigid property of the person; OntoClean forbids placing such non-

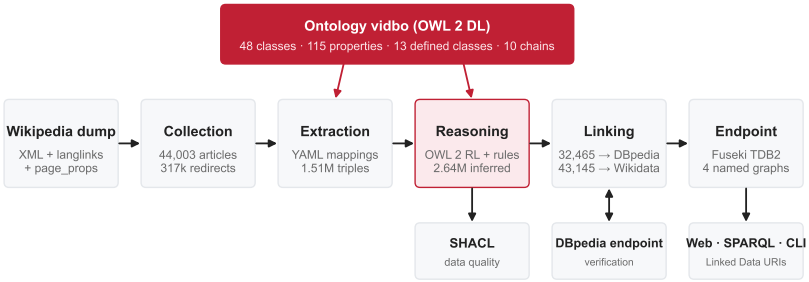


Figure 1: The ViDBpedia pipeline: dump collection, mapping-based extraction, OWL 2 RL reasoning with a rule layer, SHACL validation, linking to DBpedia and Wikidata, and publication on a Fuseki endpoint.

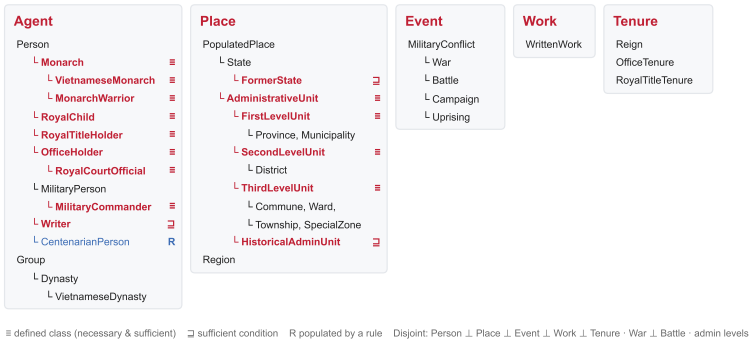


Figure 2: Class hierarchy of `vidbo`. Red classes are defined (membership is inferred by the reasoner); the blue class is populated by a SPARQL rule.

rigid classes as asserted superclasses of rigid ones [7]. We therefore never assert roles. Roles are *defined* through relations, and the reasoner classifies individuals; for instance $Monarch \equiv Person \sqcap \exists hasReign.Reign$ and $MonarchWarrior \equiv Monarch \sqcap MilitaryCommander$. A single infobox line "commander: Trần Nhân Tông" on a battle page is therefore enough to classify him as a warrior monarch. Table 1 lists the main defined classes with the number of members found on the full data.

Property axioms. Ten property chains derive relations that are never written in Wikipedia, for example

$$hasFather \circ hasFather \equiv hasPaternalGrandfather, \tag{3}$$

$$hasFather \circ memberOfDynasty \sqsubseteq memberOfDynasty, \tag{4}$$

$$bornWithin \circ locatedIn \sqsubseteq bornWithin, \tag{5}$$

$$participatedIn \circ partOfConflict \sqsubseteq participatedIn. \tag{6}$$

The chain on *memberOfDynasty* encodes patrilineal dynastic membership; the chains on *bornWithin* and *participatedIn* are right-recursive and, together with $birthPlace \sqsubseteq bornWithin$ and $commanded \sqsubseteq participatedIn$, answer "who was born in today's Nghệ An province" when an infobox only names a district, and "who took part in the Mongol–Đại Việt war" when an infobox only names a battle. Seven properties are transitive (*locatedIn*, *partOfConflict*, *hasAncestor*, *intellectualLineage*, ...), 18 pairs of properties are declared inverse, *spouse* and *sibling* are symmetric and irreflexive, *hasParent* is irreflexive and asymmetric, and *hasFather*, *birthDate* and *adminLevel* are functional. Relations that OWL cannot express are delegated to the rule layer (Section 4.3): *sibling* requires $x \neq y$, and its result feeds the chain $uncleOrAunt \equiv hasParent \circ sibling$. Table 2 gives an overview of the object properties grouped by sub-domain.

N-ary relations. "Trần Nhân Tông reigned as Emperor of Đại Việt for the Trần dynasty from 8 November 1278 to 16 April 1293" has five arguments, so following the W3C pattern [15] it becomes a *Reign* node that carries the title, the dynasty and the dates; *OfficeTenure* and

Table 1: Selected defined classes and the number of members inferred on the full data.

Class	Definition	Members
Monarch	$Person \sqcap \exists hasReign.Reign$	3,981
VietnameseMonarch	$Monarch \sqcap \exists memberOfDynasty.VietnameseDynasty$	169
RoyalChild	$Person \sqcap \exists hasParent.Monarch$	6,160
OfficeHolder	$Person \sqcap \exists holdsOffice.OfficeTenure$	6,832
RoyalCourtOfficial	$OfficeHolder \sqcap \exists servedUnder.Monarch$	204
MilitaryCommander	$Person \sqcap \exists commanded.MilitaryConflict$	10,257
MonarchWarrior	$Monarch \sqcap MilitaryCommander$	755
ThirdLevelUnit	$AdministrativeUnit \sqcap \exists adminLevel.\{3\}$	3,328

RoyalTitleTenure are analogous. The royalty infobox is shared by monarchs, consorts and heirs, all of whom have a "reign" field; titles such as *hoàng hậu* (empress consort) are mapped to *RoyalTitleTenure*, which is disjoint from *Reign*, otherwise consorts would be classified as monarchs. The chain $hasReign \circ tenureDynasty \sqsubseteq memberOfDynasty$ connects a monarch to the dynasty recorded on the reign.

The 2025 administrative reform. Administrative levels are defined by value restrictions ($FirstLevelUnit \equiv AdministrativeUnit \sqcap \exists adminLevel.\{1\}$), the domain knowledge $SecondLevelUnit \sqsubseteq HistoricalAdministrativeUnit$ records that districts were abolished on 1 July 2025, and administrative units have the key *adminCode*. A key is an OWL 2 feature that would, under OWL semantics, make two units with the same code the same individual; we use it to *report* reused codes instead (Section 6.5).

OWL 2 DL compliance. `xsd:date` and `xsd:gYear` are not in the OWL 2 datatype map, and HermiT rejects ontologies that use them as ranges. We keep these datatypes in the data, exactly like DBpedia, record the expected datatype with an annotation property, and enforce it with SHACL.

4.2 Mapping-based extraction

Collection from dumps. The MediaWiki API throttles anonymous clients to roughly 50 pages per minute, which would take about eight hours for our scope; like the DBpedia extraction framework we therefore stream the dumps (Section 3). Infoboxes are parsed with `mwparserfromhell`.

Declarative mappings. A YAML file plays the role of the DBpedia mappings wiki: 11 template groups map Vietnamese and English parameter names to `vidbo` properties, and numbered parameter groups ("*tại vị* 2", `office3`) become tenure nodes. Listing 1 shows an excerpt; the triples it produces for one article are shown, together with what the reasoner adds, in Listing 4.

Listing 1: Excerpt of the royalty mapping. Each parameter accepts its Vietnamese and English names.

```
royalty:
  names: [Thông tin nhân vật hoàng gia, Infobox royalty, ...]
  class: Person
  properties:
    "cha|father": {prop: hasFather, kind: object, max: 1}
    "sinh|birth_date": {prop: birthDate, kind: date, year_prop: birthYear}
    "miếu hiệu|temple name": {prop: templeName, kind: literal}
  tenure:                        # numbered groups -> one Reign node each
    class: Reign
    link: hasReign
    fields:
      "chức vị|succession": {prop: position, kind: literal}
      "tại vị|reign": {kind: period}
      "tiền nhiệm|predecessor": {prop: predecessor, kind: object, attach: person}
```

Vietnamese value parsers. Dates appear as "14 tháng 4, năm 2021", as templates such as `{{ngày sinh|1766|1|3}}` or `{{birth date|1258|12|7}}`, and as years BCE written "257 TCN"; periods appear as "[[8 tháng 11]] năm [[1278]] – [[16 tháng 4]] năm [[1293]]"; numbers use Vietnamese formatting ("4.412.264", "3.942,62 km²"); and lists hide links inside templates

Table 2: Main object properties of `vidbo` and their characteristics (T = transitive, S = symmetric, I = irreflexive, A = asymmetric, F = functional, IF = inverse functional).

Group	Property	Domain → range	Char.	Axioms / role in inference
Family	*hasParent*	Person → Person	I, A	$\sqsubseteq$ *hasAncestor*; $\equiv$ *hasChild*$^-$
	hasFather, *hasMother*	Person → Person	F	$\sqsubseteq$ *hasParent*; $\equiv$ `dbo:father`/`dbo:mother`
	hasAncestor	–	T	$\equiv$ *hasDescendant*$^-$
	hasGrandparent, *hasPaternalGrandfather*	–	–	chains *hasParent* $\circ$ *hasParent*, *hasFather* $\circ$ *hasFather*
	spouse, *sibling*	Person → Person	S, I	*sibling* materialised by rule R1
	uncleOrAunt	–	–	chain *hasParent* $\circ$ *sibling*
Dynasty, office	*memberOfDynasty*	Person → Dynasty	–	chains *hasFather* $\circ$ *memberOfDynasty*, *hasReign* $\circ$ *tenureDynasty*
	hasReign, *holdsOffice*, *holdsRoyalTitle*	Person → Tenure	IF	n-ary relations; define *Monarch*, *OfficeHolder*, *RoyalTitleHolder*
	servedUnder	–	–	chain *holdsOffice* $\circ$ *underMonarch*; defines *RoyalCourtOfficial*
	predecessor, *successor*	Person → Person	I	inverse pair; deliberately not transitive
Geography	*locatedIn*	Place → Place	T	$\equiv$ *contains*$^-$; $\sqsubseteq$ `dbo:isPartOf`
	province, *district*, *region*	– → admin. unit	–	$\sqsubseteq$ *locatedIn*
	seatOf	Place → Place	–	$\sqsubseteq$ *locatedIn*, $\sqsubseteq$ *capitalOf*; $\equiv$ *adminSeat*$^-$
	bornWithin, *diedWithin*	–	–	*birthPlace* $\sqsubseteq$ *bornWithin*; chain with *locatedIn*
Military	*commander*	Conflict → Person	–	$\equiv$ *commanded*$^-$; *commanded* $\sqsubseteq$ *participatedIn*
	partOfConflict	Conflict → Conflict	T	$\equiv$ *hasSubConflict*$^-$
	participatedIn	Person → Conflict	–	chain *participatedIn* $\circ$ *partOfConflict*
	conflictPlace	Conflict → Place	–	$\sqsubseteq$ *tookPlaceWithin*; chain with *locatedIn*
Works	*notableWork*	Person → Work	–	$\equiv$ *author*$^-$; $\exists$*notableWork*.*WrittenWork* $\sqsubseteq$ *Writer*
	influencedBy	Person → Person	I	$\equiv$ *influenced*$^-$; $\sqsubseteq$ *intellectualLineage* (T)

such as `{{plain list}}`. The parsers are covered by 16 unit tests. One early bug illustrates why the tests matter: in "14 tháng 4, năm 2021" the first version took the day as the year.

IRIs and provenance. Resource IRIs follow the article titles (`vidbr:Trần_Nhân_Tông`), links to redirects are resolved to canonical IRIs, and every resource records its page ID and revision as provenance.

Ontology range check. Following DBpedia, an object triple whose subject or object has an asserted class disjoint with the property's domain or range (taking super-properties and inverse properties into account) is moved into a separate dataset instead of the main graph. It removed 536 triples such as "the predecessor of Lý Thái Tổ is the Đinh dynasty" (a dynasty, not a person). Table 3 lists the resulting datasets, named as in DBpedia.

4.3 Reasoning

Why materialise. Materialising inferences lets the SPARQL endpoint answer CQs without query-time reasoning, and makes every inferred triple inspectable. Off-the-shelf pure-Python OWL RL engines were too slow for 1.5M triples, so we implemented a generic forward-chaining

Table 3: Extracted datasets. A twelfth dataset holds the 536 triples removed by the range check.

Dataset	Triples	Content
article_categories	353,687	`dct:subject`, SKOS categories
page_metadata	256,546	page ID, revision, provenance
mappingbased_literals	239,831	dates, numbers, codes, names
mappingbased_objects	206,184	entity–entity relations
tenures	135,405	reified reigns, offices and royal titles
abstracts	87,172	first paragraph of the article
wikidata_links	86,290	`owl:sameAs` to Wikidata
instance_types	45,575	types from template, title and category
labels	43,614	`rdfs:label`@vi
redirects	43,140	`wikiPageRedirects`
geo_coordinates	13,745	WGS84 latitude and longitude
Total	1,511,189	

Algorithm 1 Materialisation (OWL 2 RL rules + rule layer)

Require: ontology $\mathcal{T}$, extracted triples $\mathcal{A}$, SPARQL rules $\mathcal{R}$
compile $\mathcal{T}$: $sup^*(C)$, $supP^*(p)$, inverses, chains, $\exists/\ni$ restrictions, intersections
procedure CLOSURE($\mathcal{A}$)
  repeat
    add (s, o) to q for every $q \in supP^*(p)$ ▷ prp-spo1, prp-eqp
    add (o, s) to p^-; add (o, s) to p if p is symmetric ▷ prp-inv, prp-symp
    replace p by its transitive closure if p is transitive ▷ prp-trp
    add $p_1 \bowtie p_2$ to p for each chain $p_1 \circ p_2 \sqsubseteq p$ ▷ prp-spo2
    type subjects and objects by domains and ranges ▷ prp-dom, prp-rng
    apply $\exists p.C$ and $\exists p.\{v\}$ restrictions ▷ cls-svf1, cls-hv1/2
    add $sup^*(C)$ to every type set; add intersections ▷ cax-sco, cls-int1/2
  until no new triple
end procedure
CLOSURE($\mathcal{A}$); $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{R}(\mathcal{A})$; CLOSURE($\mathcal{A}$)
report disjointness, irreflexivity, asymmetry, functionality, key and cycle violations

reasoner that reads the axioms of `vidbo` and applies the OWL 2 RL/RDF rules [13] for class and property hierarchies, equivalence, intersection and union, `someValuesFrom` and `hasValue` restrictions, domain and range, inverse, symmetric and transitive properties and property chains (`cax-sco`, `cls-int1/2`, `cls-svf1`, `cls-hv1/2`, `prp-dom`, `prp-rng`, `prp-inv1/2`, `prp-symp`, `prp-trp`, `prp-spo1/2`) until a fix-point is reached. Nothing about the domain is hard-coded: changing the ontology changes the result.

Consistency rules are reported, not applied. The rules `cax-dw`, `prp-irp`, `prp-asyp`, the functional-property rules and `prp-key` would derive `owl:sameAs` or inconsistency. We *report* their violations instead: applying `prp-key`, for instance, would merge a new commune with the dissolved commune whose code it reuses.

Engine. Algorithm 1 outlines the engine. The TBox is compiled once into lookup tables (superclass and super-property closures, inverse pairs, chains, restrictions); the ABox is held as one set of subject–object pairs per property and one set of types per individual, so that each rule is a set operation. Transitive closure is computed per property by a graph search from every node, and a two-step chain $p_1 \circ p_2 \sqsubseteq p$ is a hash join of p_1 with p_2 indexed by subject. Rounds are naive (every rule is applied to all facts), which is simple and sufficient here: 97% of the new triples appear in the first two rounds and the remaining rounds add only the tails of long chains. Starting from 626,995 core triples (labels, categories and metadata do not take part), the fix-point is reached after 26 rounds, about 15 minutes, with 2,636,065 inferred triples.

Rule layer. Six SPARQL `CONSTRUCT` rules cover what OWL cannot express (Table 4): a different-individual condition, arithmetic and comparisons of dates. The OWL closure is computed again after the rules, so their results feed the axioms: the sibling rule R1, for instance, feeds the chain for uncles and aunts. Because the rules are plain SPARQL (Listing 2), any SPARQL engine can execute them.

Table 4: Rule layer (SPARQL `CONSTRUCT`) and the number of triples each rule derives.

Rule	Derives	Triples
R1	*sibling*: two different persons sharing a parent ($x \neq y$)	54,710
R2	*ageAtDeath* from birth and death years	10,936
R3	*CentenarianPerson* (age at death ≥ 100)	127
R4	*durationYears* of reigns and offices	22,090
R5	*populationDensity* of administrative units	5,421
R6	*reignedDuringConflict*: reign overlaps a war of the monarch's dynasty	670

Listing 2: Two rules of the rule layer: the sibling rule and the temporal rule R6.

```
# R1 - siblings: two DIFFERENT persons sharing a parent
CONSTRUCT { ?x vidbo:sibling ?y }
WHERE { ?x vidbo:hasParent ?p . ?y vidbo:hasParent ?p . FILTER (?x != ?y) }

# R6 - a monarch reigned during a conflict fought by his dynasty
CONSTRUCT { ?m vidbo:reignedDuringConflict ?c }
WHERE {
  ?m vidbo:hasReign ?r ; vidbo:memberOfDynasty ?dyn .
  ?r vidbo:startYear ?rs ; vidbo:endYear ?re .
  ?c vidbo:combatant ?dyn ; vidbo:startYear ?cs .
  FILTER (xsd:integer(STR(?rs)) <= xsd:integer(STR(?cs)) &&
          xsd:integer(STR(?cs)) <= xsd:integer(STR(?re))) }
```

4.4 Validation

OWL makes the open-world assumption: a missing value is unknown, not wrong. A qualified cardinality restriction such as "a military conflict has at least two combatants" therefore never reports a conflict with one combatant listed. SHACL [10] shapes encode the closed-world constraints we need: required labels, at most one birth date, the pattern of administrative codes (2, 3 or 5 digits), coordinate ranges and the order of start and end years. Because `sh:lessThanOrEquals` is undefined for `xsd:gYear` (including negative, BCE years), year order is checked by a SHACL-SPARQL constraint that compares integer casts (Listing 3). Validation runs over the extracted and inferred triples restricted to the 18 predicates the shapes inspect, about one million triples.

Listing 3: SHACL-SPARQL constraint checking that end years do not precede start years.

```
vidbo:YearOrderConstraint a sh:SPARQLConstraint ;
  sh:select """SELECT $this ?start ?end WHERE {
      { $this vidbo:birthYear ?start ; vidbo:deathYear ?end }
      UNION { $this vidbo:startYear ?start ; vidbo:endYear ?end }
      FILTER (xsd:integer(STR(?end)) < xsd:integer(STR(?start))) }""" .
```

4.5 Interlinking with English DBpedia

Stage 1: verified interlanguage links. The dumps contain 33,278 Vietnamese-to-English language links for our 43,614 non-disambiguation articles. A language link names an English *article*, which is not necessarily a DBpedia *resource*: the article may be newer than the DBpedia release or may have been renamed. Each candidate `dbr:` IRI is therefore verified against the public DBpedia endpoint and DBpedia redirects are followed.

Stage 2: a Silk-style matcher. The 10,930 entities without a language link are matched with a linkage rule. Candidates are retrieved by exact label lookup on label variants (the original title, the title without diacritics, and type-specific variants such as "Battle of …", "… dynasty" and "… province"). Each candidate c of an entity e receives the score

$$s(e,c) = 0.55\,\mathrm{JW}\big(\mathrm{fold}(l_e), \mathrm{fold}(l_c)\big) + 0.25\,\tau(e,c) + 0.20\,\upsilon(e,c), \tag{7}$$

where JW is the Jaro–Winkler similarity of the labels after removing diacritics and parenthesised qualifiers, $\tau \in \{0.5, 1\}$ rewards compatible types between `vidbo` and DBpedia classes, and υ rewards agreement of birth or start years (1 for equal years, 0.7 within two years, 0.2 otherwise, 0.6 if unknown). Two vetoes set $s = 0$: incompatible types, and a difference of more than two

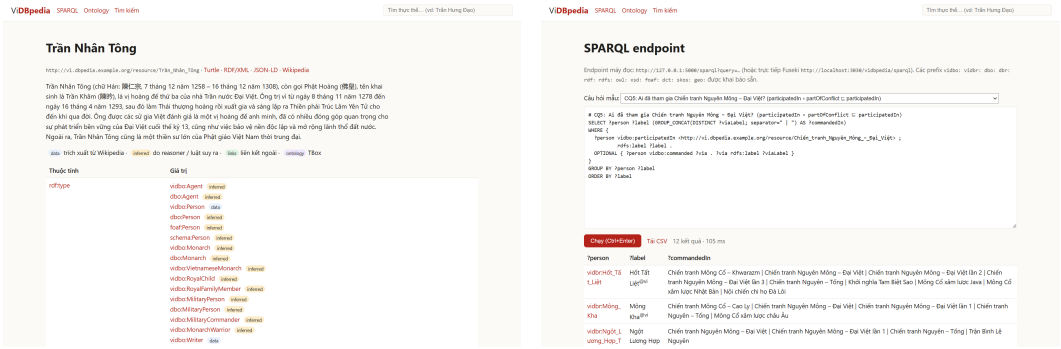


Figure 3: Left: resource page of Trần Nhân Tông, with *data* and *inferred* badges. Right: the SPARQL editor answering CQ5 in 105 ms.

Table 5: Pipeline steps, outputs and run time on a laptop.

Step (script)	Output	Time
`collect_dump.py`	44,003 articles, 317,604 redirects	13 min
`extract.py`	12 datasets, 1.51M triples	10 min
`reason.py`	2.64M inferred triples, consistency report	15 min
`validate.py`	SHACL report (104 violations)	17 min
`link.py`	32,465 DBpedia links, linking evaluation	30–60 min[a]
`load_endpoint.py`	Fuseki TDB2 store, 4 named graphs	1 min

[a]depends on the availability of the public DBpedia endpoint; answers are cached

years in the birth year of a person. A candidate is accepted if $s \geq 0.88$ and sent to manual review if $0.75 \leq s < 0.88$. The same rule is provided as a Silk link specification, and the scoring function is a dozen lines of Python in `src/link.py`.

Engineering for an unreliable endpoint. The public endpoint was unavailable several times, so the linker caches every answer and re-asks only missing queries on the next run. The current DBpedia release no longer publishes `dbo:wikiPageID` or `dbo:abstract`, so existence is tested with `rdfs:label`. Links to Wikidata come from the QIDs in the page-properties table.

4.6 Publication

The graph is loaded into an Apache Jena Fuseki [18] TDB2 store with four named graphs, *ontology*, *data*, *inferred* and *links*, whose union is the default graph, so that any triple can be traced to its origin. A web front end (Figure 3) offers DBpedia-style resource pages that badge every value as extracted, inferred or linked, a SPARQL editor preloaded with the 18 CQs, an ontology browser and a label search. Resource URIs are dereferenceable: `/resource/X` answers with HTTP 303 to an HTML page or, by content negotiation, to Turtle, RDF/XML or JSON-LD. A terminal client runs ad-hoc queries, query files or numbered CQs. Together with the CC BY-SA licence inherited from Wikipedia and the links to DBpedia and Wikidata, this meets all five Linked Data stars.

5 Experimental Setup

Protocol. All steps run on one Windows laptop with 16 GB of RAM, Python 3.13 and Java for Fuseki 5.6.0; the public DBpedia endpoint is the only external service. Table 5 lists the steps and their run time. Each step reads the output of the previous one from disk, so any step can be re-run alone. Validation is the slowest step, most of it spent in the per-node SPARQL constraint of pySHACL; reasoning keeps the whole ABox in memory as Python sets.

Competency questions. Each of the 18 CQs is a SPARQL file run against the local endpoint by `src/run_cqs.py`, which records the number of rows and the execution time and writes the results to a report. A CQ counts as answered if it returns a non-empty result that we checked by hand against Wikipedia. For each CQ we also record which axioms or rules make it answerable.

Table 6: Properties with the most inferred triples (excluding alignment copies such as `dbo:` and `foaf:` equivalents) and the mechanism that produces them.

Property	Inferred	Mechanism
hasAncestor, *hasDescendant*	618,339 each	*hasParent* $\sqsubseteq$ *hasAncestor*, transitivity, inverse
bornWithin	69,168	*birthPlace* $\sqsubseteq$ *bornWithin*, chain with *locatedIn*
sibling	54,710	rule R1, symmetry
intellectualLineage	46,305	*influencedBy* $\sqsubseteq$ *intellectualLineage*, transitivity
uncleOrAunt	42,525	chain *hasParent* $\circ$ *sibling*
participatedIn	36,702	*commanded* $\sqsubseteq$ *participatedIn*, chain with *partOfConflict*
diedWithin	23,839	*deathPlace* $\sqsubseteq$ *diedWithin*, chain with *locatedIn*
durationYears	22,090	rule R4
hasGrandparent, *hasGrandchild*	18,153 each	chain *hasParent* $\circ$ *hasParent*, inverse
commanded	15,991	inverse of *commander*
contains	15,725	inverse of *locatedIn*

Linking gold set and metrics. Stage 2 is evaluated on entities whose true link is known: we sample 1,500 entities that *do* have a verified language link, hide that link, and run the matcher. Let TP be the accepted candidates equal to the true link, FP the accepted candidates that differ from it, and n the sample size. We report

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{n}, \qquad F_1 = \frac{2PR}{P + R}, \tag{8}$$

overall and per class. Recall is measured against all sampled entities, so an entity for which the matcher finds no candidate counts as a miss.

Reproducibility. All code, mappings, rules and queries are in the project repository. After downloading the three dump files, `python run_pipeline.py` runs every step of Table 5 (`--from reason` or `--skip link` restart or skip steps), and `python src/serve.py` starts Fuseki and the web interface. Two automated tests accompany the code: `tests/test_parsers.py` (16 cases for dates, periods, numbers and links) and `tests/test_hermit.py` (HermiT on the Trần sample). `src/run_cqs.py` re-runs the 18 competency questions and writes their results to a report.

6 Results and Analysis

6.1 Inference yield

By mechanism, the 2.64M inferred triples come from transitive closure (1,318k), alignment with the DBpedia, FOAF and schema.org vocabularies (626k), class membership (276k), property chains (230k), the SPARQL rules (94k) and inverse and sub-properties (91k); Table 6 lists the properties that gain the most. Transitive closure dominates in volume, mostly *hasAncestor* and *hasDescendant* over royal genealogies, and alignment makes the graph queryable with other vocabularies; both are cheap and somewhat mechanical. The most informative additions are the 276k class memberships produced by defined classes and hierarchies, and the 230k chain results: 69k *bornWithin* links place people in the provinces and regions that contain their stated birth place, and 36.7k *participatedIn* links lift commanders of battles to the enclosing wars.

6.2 A worked example

For Trần Nhân Tông, 53 extracted triples grow to 158 inferred ones. Figure 4 shows the core of the derivation. From his infobox we know only his father and his reign; from the battle article we know that he commanded at Bạch Đằng in 1288. The reasoner adds his paternal grandfather, his membership of the Trần dynasty (through the reign and through his father), his participation in the Mongol–Đại Việt war (through the chain over *partOfConflict*), and four classes: *VietnameseMonarch*, *MonarchWarrior*, *MilitaryCommander* and *RoyalChild*. The full description served by the endpoint is excerpted in Listing 4.

Listing 4: Description of Trần Nhân Tông in ViDBpedia (excerpt), grouped by named graph.

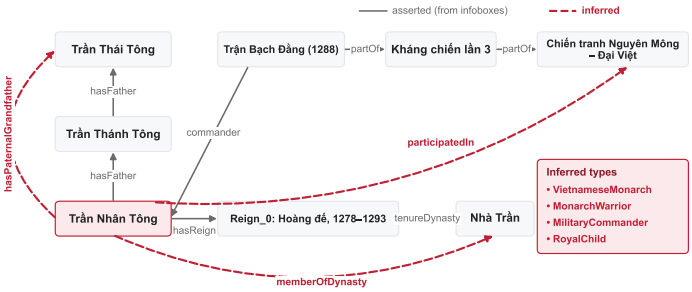


Figure 4: Asserted infobox facts (grey) and facts inferred by the ontology (red, dashed) for Trần Nhân Tông.

```
# graph <.../graph/data>  (extracted from the infobox)
vidbr:Trần_Nhân_Tông a vidbo:Person, vidbo:NationalHero ;
  vidbo:hasFather vidbr:Trần_Thánh_Tông ;
  vidbo:birthPlace vidbr:Thăng_Long ;
  vidbo:hasReign vidbr:Trần_Nhân_Tông__Reign_0 .
# graph <.../graph/inferred>  (reasoner + rules)
vidbr:Trần_Nhân_Tông a vidbo:Monarch, vidbo:VietnameseMonarch,
    vidbo:MonarchWarrior, vidbo:RoyalChild, dbo:Person, foaf:Person ;
  vidbo:hasPaternalGrandfather vidbr:Trần_Thái_Tông ;
  vidbo:participatedIn vidbr:Chiến_tranh_Nguyên_Mông_-_Đại_Việt ;
  vidbo:bornWithin vidbr:Đại_Việt ;
  vidbo:sibling vidbr:Thiên_Thụy_Công_chúa ;     # rule R1
  vidbo:uncleOrAunt vidbr:Trần_Quang_Khải ;      # chain over R1
  vidbo:ageAtDeath 50 .                           # rule R2
# graph <.../graph/links>
vidbr:Trần_Nhân_Tông owl:sameAs dbr:Trần_Nhân_Tông , wd:Q72083 .
```

6.3 Competency questions

All 18 CQs return answers on the local endpoint; the median execution time is 44 ms, with a maximum of 2.1 s for the coverage statistics and 1.7 s for the federated CQ13, which calls the public DBpedia endpoint. Table 7 lists the 18 CQs with the mechanism that makes them answerable. Every CQ relies on at least one inferred triple (a defined class, a chain, an inverse or a rule result), and CQ2 is answered entirely from the *inferred* graph. Listing 5 shows two of the formal CQs: CQ8 asks only for the inferred property *bornWithin*, and CQ13 joins local data with the public DBpedia endpoint through `owl:sameAs`.

Listing 5: Two formal competency questions: CQ8 (inferred property only) and CQ13 (federated with DBpedia).

```
# CQ8 - people born within today's Nghệ An
SELECT ?person ?label ?birthYear WHERE {
  ?person vidbo:bornWithin vidbr:Nghệ_An ; rdfs:label ?label .
  OPTIONAL { ?person vidbo:birthYear ?birthYear } }

# CQ13 - federated: English descriptions from DBpedia
SELECT ?label ?dbpedia ?descriptionEn WHERE {
  SERVICE <https://dbpedia.org/sparql> {
    ?dbpedia dct:subject dbc:Vietnamese_monarchs .
    OPTIONAL { ?dbpedia dbo:description ?descriptionEn
               FILTER (LANG(?descriptionEn) = "en") } }
  ?m owl:sameAs ?dbpedia ; vidbo:memberOfDynasty vidbr:Nhà_Lý ;
     a vidbo:Monarch ; rdfs:label ?label . }
```

6.4 Verification with a tableau reasoner

On the Trần sample, HermiT reports no unsatisfiable class and derives all eleven expected facts: the three emperors are *VietnameseMonarch*, Trần Nhân Tông is also *MonarchWarrior* and *RoyalChild*, Trần Hưng Đạo is *MilitaryCommander*, and the chains yield *memberOfDynasty*, *hasPaternalGrandfather*, *participatedIn*, *bornWithin* and *tookPlaceWithin*. At the schema level HermiT additionally derives *adminSeat* $\sqsubseteq$ *contains* from *seatOf* $\sqsubseteq$ *locatedIn* and the two in-

Table 7: The 18 competency questions: mechanism, number of rows and execution time on the local Fuseki endpoint.

CQ	Question	Mechanism (axioms / rules involved)	Rows	ms
1	Vietnamese monarchs by dynasty and accession year	defined classes *Monarch*, *VietnameseMonarch*; chain *hasReign* ∘ *tenureDynasty*	189	246
2	Paternal grandfather of Trần Nhân Tông	chain *hasFather* ∘ *hasFather*	1	11
3	Descendants of Trần Thái Tông	*hasParent* ⊑ *hasAncestor* (T), inverse *hasDescendant*	62	30
4	Vietnamese monarchs who commanded battles	*MonarchWarrior* ≡ *Monarch* ⊓ *MilitaryCommander*, inverse *commanded*	89	47
5	Participants of the Mongol–Đại Việt war	chain *participatedIn* ∘ *partOfConflict*, *partOfConflict* (T)	12	37
6	Reigns of the Trần dynasty with their length	n-ary *Reign* nodes, rule R4	29	27
7	Conflicts per present-day province	chain *tookPlaceWithin* ∘ *locatedIn*, *locatedIn* (T)	30	117
8	People born within today's Nghệ An	*birthPlace* ⊑ *bornWithin*, chain with *locatedIn*	264	116
9	Provinces after the 2025 merger: communes, population, density	*ThirdLevelUnit* by `hasValue`, *province* ⊑ *locatedIn*, rule R5	28	110
10	Monarchs reigning while their dynasty fought a conflict	temporal rule R6	166	30
11	Siblings, uncles/aunts and grandparents of Trần Nhân Tông	rule R1, chains *uncleOrAunt*, *hasGrandparent*	20	34
12	People who lived to 100 or more	rules R2, R3	50	41
13	English descriptions of Lý-dynasty monarchs from DBpedia	`owl:sameAs`, SPARQL `SERVICE`	9	1,698
14	Monarchs with the most court officials	chain *holdsOffice* ∘ *underMonarch*, *RoyalCourtOfficial*	15	39
15	Writers influenced (directly or not) by Nguyễn Du	inverse *influenced*, *intellectualLineage* (T)	2	10
16	Districts abolished on 1 July 2025, per province	*SecondLevelUnit* ⊑ *HistoricalAdministrativeUnit*	65	48
17	Data errors (death before birth, *Person* and *Place*)	disjointness, comparison of years	100	417
18	Share of entities linked to DBpedia, by class	`owl:sameAs`	6	2,051

verse pairs, an entailment our engine does not need because it works on the data. The same facts are produced by our RL engine on the full data, which cross-checks the implementation.

6.5 Data quality

Table 8 summarises the issues detected by the axioms and the shapes, about 1,700 in total. All examples are genuine errors or ambiguities in Vietnamese Wikipedia; the article on Anna Freud, for instance, gives 1985 as her year of birth (it should be 1895), and the infobox of Lý Thái Tổ names a dynasty as his predecessor. The transitive cycles are the largest group: most are reversed "influenced by" fields between writers and pairs such as Minsk and Minsk Region, each claiming to lie within the other.

6.6 Linking

Stage 1. 32,136 of the 33,278 language links are verified on DBpedia (937 after a redirect); 1,142 point to articles missing from the DBpedia release.

Stage 2. On the 1,500-entity gold sample, the matcher reaches a precision of 96.4% (728 correct, 27 wrong), a recall of 48.5% and an F1 of 0.646 at the acceptance threshold. Applied to the 10,930 entities without a language link, it accepts 329 links and sends 319 candidates to review. In total

Table 8: Issues detected by axioms and shapes, with an example of each.

Check	Found	Example
Range check at extraction	536	predecessor of Lý Thái Tổ = Đinh dynasty (a dynasty, not a person)
Disjoint classes	145	a "father" field linking to a country ($Person \sqcap Place \sqsubseteq \bot$)
Cycles of transitive properties	857	Shakespeare and Henry James influencing each other
Asymmetric *hasParent*	6	father and son swapped (Jesse Root Grant, Ulysses S. Grant)
Functional *birthDate*	6	two birth years for Alexandre de Rhodes (1591, 1593)
`hasKey`(*adminCode*)	19	2025 communes reusing the code of a dissolved commune
SHACL (closed world)	104	Anna Freud: born 1985, died 1982

Table 9: Stage-2 matcher on the 1,500-entity gold sample, by class.

Class	*n*	TP	FP	Precision	Recall
Person	1,127	637	12	98.2%	56.5%
Military conflict	126	0	0	–	0.0%
State	91	9	2	81.8%	9.9%
Place	82	52	5	91.2%	63.4%
Administrative unit	67	29	8	78.4%	43.3%
Other	7	1	0	100%	14.3%
All	1,500	728	27	96.4%	48.5%

ViDBpedia has 32,465 `owl:sameAs` links to DBpedia and 43,145 to Wikidata. Table 9 breaks the evaluation down by class and Figure 5 shows the resulting coverage.

Error analysis. Persons, three quarters of the sample, are matched with 98.2% precision; dates are a strong signal for them, and the year veto removes most homonyms. Administrative units are the hardest (78.4%): the 2025 reform created communes and wards that share names with former districts, and the matcher prefers the shorter title, e.g. "Trà Cú (huyện)" was matched to `dbr:Trà_Cú` instead of `dbr:Trà_Cú_district`. Other errors cross types that DBpedia leaves untyped: the Vietnamese article on the destroyer *USS Mannert L. Abele* was matched to the person of the same name. For military conflicts the matcher proposes no candidate at all: Vietnamese battle titles rarely translate literally into the English titles used by DBpedia, which is a limitation of label-based blocking.

6.7 Summary of key findings

1. **The ontology adds more than was extracted.** Reasoning adds 2.64M triples to 627k core extracted ones; 276k class memberships and 230k chain results are new, informative facts.
2. **Every competency question needs inference.** All 18 CQs are answered (median 44 ms), and none of them could be answered from the extracted triples alone.
3. **Axioms double as a quality checker.** Disjointness, transitivity, functionality, keys and SHACL expose about 1,700 issues in Wikipedia, and disjointness also exposed a bug in our own extractor.
4. **Linking is precise but incomplete.** 96.4% precision and 48.5% recall on 1,500 gold links; persons link well, conflicts not at all.

7 Discussion and Limitations

Axioms are inference rules, not constraints. The most important lesson of the project is that a domain, range or transitivity axiom is not documentation: the reasoner applies it to every triple. Three design errors were caught only because the reasoner amplified them. (i) A first version declared *predecessor* transitive; since infoboxes list predecessors per office and a person holds many offices, the closure connected almost all office holders and produced 3.8M meaningless triples. (ii) The domain of *posthumousName*, *templeName* and *eraName* was *Monarch*; empresses and princes also receive posthumous names, so 925 of them were inferred to be monarchs until the domain was relaxed to *Person*. (iii) Regional infoboxes (e.g. the Red River Delta)

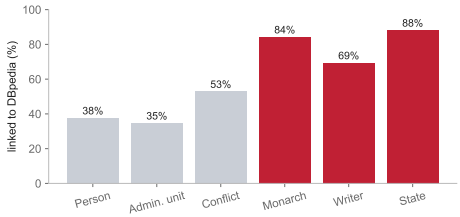


Figure 5: Share of entities linked to English DBpedia, by class. States and monarchs are well linked; most persons and the new administrative units have no English counterpart.

list their provinces; reading these lists as *locatedIn* created containment cycles, and transitivity placed a border campaign in a dozen provinces. CQ7 exposed the error, and the lists are now read as *contains*. Because transitive properties are non-simple, OWL 2 DL cannot declare them irreflexive; we therefore detect cycles explicitly.

Disjointness catches software bugs. In one run, the axiom $\mathit{Person} \sqcap \mathit{Tenure} \sqsubseteq \bot$ reported 17,841 violations. The cause was a YAML mapping key `on:` that the YAML 1.1 parser read as the boolean `true`, which attached predecessors to tenure nodes; the domain of *predecessor* then typed those nodes as persons. No unit test would have found this: the extractor ran without error and produced plausible triples.

Practical recommendations. (1) Keep the competency questions as executable queries; they decide which axioms are worth their reasoning cost. (2) Treat every new axiom as a code change and compare the inferred counts per property after re-running the reasoner. (3) Use disjointness generously: it adds no facts but is the cheapest error detector.

Threats to validity. The competency questions were written by us, so their success measures whether the ontology does what we designed it to do, not how well it serves other users. The linking gold set is drawn from entities that have an English article, which are on average better known than those without one; recall on the entities that actually lack a link is likely lower than 48.5%.

Limitations and future work. ViDBpedia covers a domain (44k of 1.6M articles) rather than the whole Vietnamese Wikipedia; it uses a placeholder namespace (`vi.dbpedia.example.org`); lunar dates and era names are kept as text; and stage-2 linking cannot match conflicts. Materialisation favours query speed over freshness: an update requires re-running the reasoner. Future work includes extending the mappings to the whole Vietnamese Wikipedia, normalising lunar dates and era names, contributing the mappings to the DBpedia community, and learning the linkage rule from the gold pairs to improve recall.

8 Conclusion

ViDBpedia shows that a Vietnamese DBpedia can be more than a translation of infoboxes into RDF. An ontology designed from competency questions, with defined classes, property chains, reified tenures and explicit knowledge about the 2025 reform, lets a simple OWL 2 RL engine add 2.64M triples to the extracted ones and answer all 18 competency questions, while the same axioms and a set of SHACL shapes surface about 1,700 data-quality issues. The dataset meets the five Linked Data stars, with 32,465 links to DBpedia at 96.4% matcher precision. Beyond the specific dataset, the project yields one transferable lesson: derive the axioms from the questions, then let the reasoner both answer the questions and audit the data.

Contributions of team members. All four authors contributed equally to the project; Table 10 gives the per-member breakdown.

References

[1] S. Auer, C. Bizer, G. Kobilarov, J. Lehmann, R. Cyganiak, and Z. Ives. DBpedia: A nucleus for a web of open data. In *The Semantic Web: 6th International Semantic Web Conference (ISWC 2007)*, volume 4825 of *Lecture Notes in Computer Science*, pages 722–735. Springer, 2007.

Table 10: Contributions of team members.

Member	Code / module	Writing & analysis
Duong Hoang Hai (20252273M)	Dump collection, infobox mappings, value parsers, OWL 2 RL reasoner, rules, SHACL	§1 Introduction; §3 Data; §4.2–4.4; §6.5 Data quality
Nguyen Thi Mai Anh (20252265M)	Ontology `vidbo`, competency questions, HermiT verification, system demo	§2 Related work; §4.1 Ontology; §6.2, §6.4
Le Phu Tai (20252268M)	DBpedia / Wikidata linking, Silk-style matcher, Fuseki endpoint, web interface, CLI	§4.5 Interlinking; §4.6 Publication; §6.6 Linking
Le Van Luan (20252572M)	Competency-question runner, evaluation of results and data quality	Abstract; §5; §6.1, 6.3, 6.7; §7–8

[2] T. Berners-Lee. Linked data – design issues. W3C, `https://www.w3.org/DesignIssues/LinkedData.html`, 2006.

[3] C. Bizer, J. Lehmann, G. Kobilarov, S. Auer, C. Becker, R. Cyganiak, and S. Hellmann. DBpedia – a crystallization point for the web of data. *Journal of Web Semantics*, 7(3):154–165, 2009.

[4] A. Gangemi. Ontology design patterns for semantic web content. In *The Semantic Web – ISWC 2005*, volume 3729 of *Lecture Notes in Computer Science*, pages 262–276. Springer, 2005.

[5] B. Glimm, I. Horrocks, B. Motik, G. Stoilos, and Z. Wang. HermiT: An OWL 2 reasoner. *Journal of Automated Reasoning*, 53(3):245–269, 2014.

[6] M. Grüninger and M. S. Fox. Methodology for the design and evaluation of ontologies. In *IJCAI-95 Workshop on Basic Ontological Issues in Knowledge Sharing*, 1995.

[7] N. Guarino and C. Welty. Evaluating ontological decisions with OntoClean. *Communications of the ACM*, 45(2):61–65, 2002.

[8] S. Harris and A. Seaborne. SPARQL 1.1 query language. W3C Recommendation, `https://www.w3.org/TR/sparql11-query/`, 2013.

[9] T. Heath and C. Bizer. *Linked Data: Evolving the Web into a Global Data Space*. Morgan & Claypool, 2011.

[10] H. Knublauch and D. Kontokostas. Shapes constraint language (SHACL). W3C Recommendation, `https://www.w3.org/TR/shacl/`, 2017.

[11] D. Kontokostas, C. Bratsas, S. Auer, S. Hellmann, I. Antoniou, and G. Metakides. Internationalization of linked data: The case of the Greek DBpedia edition. *Journal of Web Semantics*, 15:51–61, 2012.

[12] J. Lehmann, R. Isele, M. Jakob, A. Jentzsch, D. Kontokostas, P. N. Mendes, S. Hellmann, M. Morsey, P. van Kleef, S. Auer, and C. Bizer. DBpedia – a large-scale, multilingual knowledge base extracted from Wikipedia. *Semantic Web*, 6(2):167–195, 2015.

[13] B. Motik, B. C. Grau, I. Horrocks, Z. Wu, A. Fokoue, and C. Lutz. OWL 2 web ontology language profiles (second edition). W3C Recommendation, `https://www.w3.org/TR/owl2-profiles/`, 2012.

[14] National Assembly of Vietnam. Resolution no. 202/2025/QH15 on the arrangement of provincial-level administrative units. Adopted on 12 June 2025, 2025.

[15] N. Noy and A. Rector. Defining n-ary relations on the semantic web. W3C Working Group Note, `https://www.w3.org/TR/swbp-n-aryRelations/`, 2006.

[16] H. Paulheim. Knowledge graph refinement: A survey of approaches and evaluation methods. *Semantic Web*, 8(3):489–508, 2017.

[17] F. M. Suchanek, G. Kasneci, and G. Weikum. YAGO: A core of semantic knowledge. In *Proceedings of the 16th International Conference on World Wide Web (WWW 2007)*, pages 697–706, 2007.

[18] The Apache Software Foundation. Apache Jena and Fuseki. `https://jena.apache.org/`, 2026. Version 5.6.0.

[19] J. Volz, C. Bizer, M. Gaedke, and G. Kobilarov. Silk – a link discovery framework for the web of data. In *Proceedings of the WWW2009 Workshop on Linked Data on the Web (LDOW 2009)*, 2009.

[20] D. Vrandečić and M. Krötzsch. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85, 2014.

[21] W3C OWL Working Group. OWL 2 web ontology language document overview (second edition). W3C Recommendation, `https://www.w3.org/TR/owl2-overview/`, 2012.

[22] A. Zaveri, A. Rula, A. Maurino, R. Pietrobon, J. Lehmann, and S. Auer. Quality assessment for linked data: A survey. *Semantic Web*, 7(1):63–93, 2016.